Progress Report — Non-Invasive Optical Glucose Monitoring Prototype

Project: i-Lab Optical Glucose Monitoring Project
Focus: Optical signal acquisition, Arduino–MATLAB communication, preliminary red-light measurements, and signal-processing workflow
Status: Initial data-acquisition pipeline established; signal quality and physiological interpretation remain under evaluation.

1. Objective

The objective of this session was to establish a repeatable data-acquisition workflow for the non-invasive optical glucose monitoring prototype. The immediate focus was to acquire photodiode measurements under red LED illumination, transfer the data to MATLAB, preserve the raw measurements, and begin evaluating whether the recorded optical signal contains a distinguishable pulsatile component suitable for further analysis.

At this stage, the work is focused on validating the sensing and data-processing pipeline rather than estimating blood glucose concentration.

2. Hardware and Data-Acquisition Setup

The prototype uses an optical sensing arrangement consisting of:

- **Red LED:** 660 nm optical illumination.
- **Infrared LED:** 940 nm optical illumination.
- **Photodetector:** BPW34 photodiode.
- **Analog front end:** Transimpedance amplifier (TIA) circuit for converting the photodiode current into a measurable voltage.
- **Microcontroller:** Arduino Uno.
- **Data-processing environment:** MATLAB R2025b.

The Arduino configuration used during the session was:

Component	Configuration
Red LED control	Digital pin 10
Infrared LED control	Digital pin 11
Analog sensor input	A0
Serial communication	115200 baud
Sampling interval	Approximately 10 ms
Target sampling frequency	Approximately 100 Hz

The LEDs were controlled individually so that measurements could be collected under a specific illumination condition. For the red-light acquisition, the red LED was switched on while the infrared LED remained off.

The nominal sampling frequency was approximately 100 Hz, based on a 10 ms sampling interval. This is the intended acquisition rate; the actual rate and timing consistency should be verified from timestamps or acquisition diagnostics.

3. Electronics and Initial Troubleshooting

During the session, a power-supply configuration issue was identified: the TIA/op-amp circuit had been powered at 3.3 V instead of the intended 5 V supply.

This was an important troubleshooting finding because the analog front end directly affects the signal delivered to the Arduino. An incorrect supply voltage can influence amplifier operation, available output range, and the interpretation of measured values.

The supply configuration should therefore be verified before comparing subsequent measurements. The amplifier's permitted supply range and the circuit's output limits must also be checked against the relevant component datasheets.

Validation requirement: The available results do not, by themselves, establish how much this supply issue affected the collected readings. Measurements obtained under different circuit configurations should be documented separately rather than treated as directly comparable.

Feedback Component Adjustment in the Transimpedance Amplifier

During the session, the feedback components of the transimpedance amplifier (TIA) circuit were changed to the following values:

Component	Updated value
Feedback resistor (R_f))	68 kΩ
Feedback capacitor (C_f))	100 pF

The feedback resistor converts the photodiode's output current into a voltage that can be measured by the Arduino's analog input. Its value therefore influences the transimpedance gain and the resulting output-voltage range.

The feedback capacitor affects the amplifier's frequency response and stability. Depending on the photodiode capacitance, amplifier characteristics, and circuit layout, it can help control high-frequency noise and reduce the risk of instability.

These component values were updated as part of the effort to improve the analog front end for optical signal acquisition. Further measurements are required to determine whether the revised

configuration improves signal stability, noise performance, and the visibility of any physiological pulsatile component.

The updated component values should be retained in the circuit documentation and used as the reference configuration for subsequent tests, unless further adjustments are required.

4. Arduino–MATLAB Communication

A serial acquisition workflow was established between the Arduino and MATLAB.

The Arduino periodically reads the analog input and transmits the ADC value through the serial connection at 115200 baud. MATLAB receives and parses the incoming numerical values, enabling the signal to be plotted and stored for further analysis.

The workflow demonstrated the feasibility of transferring sensor readings into MATLAB for inspection and processing.

The principal data-acquisition requirements established during the session were:

1. Control the selected LED independently.
2. Read the photo detector output through analog input A0.
3. Transmit numerical readings over serial communication.
4. Receive the data in MATLAB.
5. Visualize the signal during or after acquisition.
6. Preserve raw readings for subsequent analysis.

Keeping the raw data available is important because filtering and baseline removal can change the appearance of a signal. Raw measurements should remain the primary record, while processed signals should be treated as additional analytical outputs.

5. Preliminary Red-Light Measurements

A red-light measurement run was collected and examined. The readings varied over time, with an observed baseline trend and smaller fluctuations superimposed on the overall signal.

These observations indicate that the recorded signal is not simply a constant value. However, variation alone does not confirm that the signal contains a reliable photoplethysmography (PPG) waveform.

Potential contributors to the observed variation include:

- Changes in optical coupling or finger positioning.
- Ambient-light interference.

- Baseline drift in the analog front end.
- Electrical noise or supply-related effects.
- Changes in tissue blood volume associated with the cardiac cycle.
- Sampling or LED-switching effects, where applicable.

These are possible explanations rather than confirmed causes. Further controlled measurements are required to distinguish physiological fluctuations from instrumental effects.

Interpretation

The red-light measurements provide an initial dataset for characterizing the optical sensing system. At this stage, the presence, periodicity, and stability of a physiological PPG component have not been conclusively established.

No glucose concentration should be inferred from these readings without an independently validated calibration method and suitable reference measurements.

6. MATLAB Signal-Processing Workflow

The MATLAB workflow was developed around preserving the original signal and examining processed versions separately.

6.1 Raw signal visualization

The raw ADC readings are plotted against time to inspect the signal's overall behavior, including baseline drift, abrupt transitions, and short-term fluctuations.

6.2 Baseline estimation and AC extraction

A moving-average baseline can be estimated and subtracted from the raw signal:

$$AC(t) = x(t) - DC_{\text{estimated}}(t)$$

where (x(t)) is the raw signal and (DC estimated (t)) is the slowly varying baseline estimate.

This provides a preliminary way to examine fluctuations around the baseline. However, the resulting AC component is not automatically a valid PPG signal; it may still contain noise and other non-physiological variations.

6.3 Band-pass filtering

A Butterworth band-pass filter with an initial frequency range of approximately 0.5–5 Hz was considered for examining potential pulse-related fluctuations.

This range is a starting point for signal inspection, not a finalized filter specification. The filter should be applied only after verifying the sampling frequency and assessing the signal characteristics. Zero-phase filtering with MATLAB's `filtfilt` function can be used for offline analysis when appropriate.

The raw signal, baseline-subtracted signal, and filtered signal should be plotted separately to make the effect of each processing step transparent.

6.4 Data preservation and run organization

A daily master CSV workflow was planned so that individual acquisition runs could be preserved and compared without overwriting previous measurements.

The intended organization is one column per run, with a consistent date-based naming convention. Because different runs may contain different numbers of samples, shorter columns should be padded with `NaN` when necessary, rather than discarding valid readings.

The master dataset should contain the original readings. Filtered data and derived features can be stored separately or exported to dedicated files.

7. Progress Achieved

The session established several important elements of the experimental workflow:

- An Arduino-based optical data-acquisition configuration was established.
- Individual control of the red and infrared LEDs was incorporated into the measurement approach.
- Serial transmission of numerical sensor readings to MATLAB was demonstrated.
- MATLAB visualization of incoming or collected measurements was established.
- Initial red-light measurements were collected for signal characterization.
- Baseline removal and band-pass filtering were identified as candidate processing steps.
- A repeatable, date-based workflow for preserving individual measurement runs was planned.
- A TIA/op-amp power-supply configuration issue was identified for further verification.

These developments establish an initial experimental platform for continued investigation. They do not yet demonstrate validated PPG extraction or non-invasive glucose estimation.

8. Current Limitations

The following limitations remain:

1. **Signal quality:** A clear and repeatable PPG waveform has not yet been confirmed.
2. **Baseline drift:** The source and magnitude of baseline variation need further investigation.

3. **Analog front-end validation:** The TIA supply configuration and output behavior require verification.
4. **Sampling validation:** The nominal 100 Hz sampling rate should be checked against actual acquisition timing.
5. **Measurement repeatability:** Multiple controlled runs are needed to assess consistency.
6. **Optical conditions:** LED positioning, sensor alignment, ambient light, and finger pressure should be controlled.
7. **Red versus infrared comparison:** Measurements from both wavelengths must be acquired under documented and comparable conditions.
8. **Glucose calibration:** No validated relationship between the current ADC readings and reference glucose concentrations has been established.

9. Planned Next Steps

Step 1 — Verify the hardware

- Confirm the TIA/op-amp supply voltage and wiring.
- Check amplifier output behavior and Arduino ADC limits.
- Verify LED control and ensure the red and infrared LEDs operate independently.
- Check the photodetector response under controlled illumination and ambient-light conditions.

Step 2 — Repeat controlled acquisitions

- Collect separate red-light and infrared-light runs.
- Keep sensor positioning and measurement conditions as consistent as possible.
- Record the run identifier, wavelength, duration, sampling configuration, and relevant experimental notes.
- Repeat measurements to assess signal stability.

Step 3 — Validate acquisition timing

- Confirm the actual sampling frequency and sample count.
- Inspect missing, duplicated, or irregular samples.
- Use timestamps or acquisition diagnostics if the nominal 10 ms interval is insufficient to characterize timing accurately.

Step 4 — Analyze the signals in MATLAB

- Plot the raw signal.
- Estimate and inspect baseline drift.
- Extract the AC component.
- Apply an appropriate band-pass filter for exploratory PPG analysis.
- Inspect periodicity and potential pulse peaks.

- Compare raw and processed signals to avoid interpreting noise or switching artifacts as physiological activity.

Step 5 — Organize and document the data

- Maintain the daily master CSV without overwriting previous runs.
- Keep raw data separate from processed outputs.
- Document the experimental conditions associated with every run.
- Upload the acquisition code, MATLAB analysis scripts, and concise experimental notes to GitHub.

Step 6 — Establish a foundation for glucose-related analysis

Only after the optical signal and measurement process have been characterized should the project proceed toward comparing wavelengths, extracting candidate features, and evaluating calibration approaches against independent reference measurements.

10. Conclusion

The last session focused on establishing the basic experimental and computational workflow for the optical glucose-monitoring prototype. Arduino-to-MATLAB data transfer and preliminary red-light acquisition provided an initial dataset for investigating the optical signal.

The session also highlighted the importance of analog front-end configuration, baseline stability, acquisition timing, and systematic data preservation. The immediate priority is to verify the hardware, acquire comparable red and infrared measurements, and determine whether a reproducible physiological signal can be extracted.

Overall status: Initial data-acquisition workflow established; optical signal characterization and validation remain ongoing.